TOPOLOKA: Master Submission Document and Supplementary Materials

Techfest 2026-27, IIT Bombay - Namma Space Competition (Official Competition Deliverable)

Submission Details	Verification & Access
Author / Lead	**Shashank Dev** (`coderperson043@gmail.com`)
Team ID	**TM-60A7F03511D**
Competition ID	**NAMM-B3B981411E93**
Live Free-Roam Web Twin	https://sciencebanda09.github.io/topoloka/
Source Code Repository	https://github.com/sciencebanda09/topoloka (MIT License)
Demo Walkthrough Video	https://github.com/sciencebanda09/topoloka/releases/download/r1-demo/topoloka_demo.mp4
Technical Abstract (PDF)	`submission/Technical_Abstract_Topoloka.pdf` (4 pages, IEEE/ACM format)
Kaggle Cloud Pipelines	`kaggle/kernels/` ($0.00 compute expenditure)
Local Merged Video	`topoloka_demo.mp4` (Full HD 1080p, 30 FPS, H.264 + AAC)

1. Master Links and Supplementary Drive Index

All project deliverables, interactive web demonstrators, video recordings, and source code are consolidated in the master index below:

Deliverable	Resource / Access URL	Description
Live Free-Roam Web Twin	https://sciencebanda09.github.io/topoloka/	WebGL 3DGS viewer. Default opens **Studio Room** with in-viewer orientation controls; dropdown allows switching to **Melbourne Metro**. Runs at 60 FPS in standard browsers.
Source Code Repository	https://github.com/sciencebanda09/topoloka	Complete end-to-end codebase: smartphone capture SOP, SfM pipeline, depth metricization, 3DGS export, and Vite/TypeScript WebGL client. MIT Open Source.
Demo Walkthrough Video	topoloka_demo.mp4	High-definition (1080p, 30 FPS, H.264+AAC) walkthrough combining Metro corridor 3DGS roaming, multi-representation point cloud inspection, and Studio Room live HUD orientation controls.
Technical Abstract (PDF)	`submission/Technical_Abstract_Topoloka.pdf`	Formal 4-page academic report detailing mathematical formulation, pipeline architecture, and empirical benchmarks.
Kaggle Cloud Pipelines	`kaggle/kernels/`	Automated GPU scripts executing frame extraction, pycolmap SfM, Depth Anything V2 metric alignment, and splatfacto training ($0.00 total compute cost).
Supplementary Artifacts	`web/public/artifacts/`	Pre-computed Gaussian splat binaries (`splat.splat`), camera trajectory matrices (`nav.json`), and metadata logs (`meta.json`).

2. Executive Summary

Overview

Topoloka (*topos* [place] + *loka* [world]) is a spatial intelligence engine that converts a standard smartphone walk-through video into an interactive, metric 3D digital twin delivered as a lightweight, zero-install web link.

Unlike traditional photogrammetry or laser scanning pipelines that require expensive proprietary LiDAR hardware, Topoloka achieves sub-decimeter metric fidelity directly from consumer video. The resulting digital twin delivers free first-person roaming at 50-60 FPS on standard integrated laptop GPUs and mobile phones, featuring obstacle-aware pathfinding and live in-viewer spatial orientation controls.

The Problem

- **LiDAR Hardware Barrier:** Professional indoor mapping platforms (e.g., Matterport Pro3, Leica BLK360) cost between $4,000 and $25,000, limiting spatial digitalization to well-funded enterprises.
- **Heavy Client Payloads:** Standard mesh reconstructions with high-resolution texture atlases regularly exceed 300 MB to 800 MB, causing excessive load times, severe GPU texture thrashing, and unplayable mobile frame rates.
- **Lack of Traversability and Navigation:** Conventional NeRF / 3DGS viewers function solely as passive camera visualizers; they lack physical floor collision, walkable boundaries, and navigation graphs.

The Topoloka Solution

- **Consumer Monocular Video Input:** Uses ordinary 4K/1080p video recorded on any modern smartphone following our simple capture Standard Operating Procedure (SOP).

- **Metric Scaling via Foundation AI:** Integrates Depth Anything V2 Metric Indoor Large to infer true real-world scale (meters per scene unit) without physical scale markers or hardware depth sensors.
- **Sub-50 MB Static Web Payload:** Delivers full 30,000-iteration studio-grade 3D Gaussian Splats (1.59M Gaussians) over static web hosting (GitHub Pages) with initial cold loads in under 22 seconds.
- **Built-in 2.5D Obstacle Costmap and A* Navigation:** Simultaneously outputs a metric 5 cm occupancy costmap that prevents users from walking through walls and furniture, computing real-time waypoints to spatial goals.
- **Live In-Viewer Orientation Controls:** Features an interactive HUD toolbar allowing visitors to rotate, tilt, roll, flip 180 degrees, and level scenes live inside the browser, automatically saving custom camera poses to shareable URL parameters (`?rx=&ry=&rz=&oy=`).

3. Measured Benchmarks and Empirical Performance

All metrics below are measured directly on the deployed GitHub Pages release running on consumer hardware:

Performance Metric	Studio Room Twin (`room`)	Melbourne Metro Benchmark (`metro_v1`)
Gaussian Splats	**1,593,376** (30,000 iterations)	**457,221** (25,000 iterations)
Walkable Area	**73.30 m²**	**315.05 m²**
Registration Rate	100% full coverage	216 / 216 frames (100.0%)
Reprojection Error	Sub-pixel	**0.403 px median** (0.613 px mean)
Metric Scale Accuracy	Calibrated indoor benchmark	**6.2505 m/unit (7.54% MAD)**
Total Artifact Payload	48.6 MB (Binary Splat + Nav)	18.4 MB (Splat + Cloud + Nav)
Cold Load Time	<22 seconds (50 Mbps)	**19.2 seconds** (50 Mbps)
Desktop Frame Rate	**60.0 FPS** (Chrome / Edge / ANGLE)	**60.1 FPS** (Chrome / Edge / ANGLE)
Integrated iGPU FPS	**48 to 55 FPS** (AMD Radeon 780M)	**51.0 to 60.1 FPS** (AMD Radeon 780M)
Mobile Web FPS	**60.0 FPS** (WebKit / Chrome Mobile)	**59.9 FPS** (WebKit / Chrome Mobile)
Total Cloud GPU Cost	**$0.00** (Kaggle Free Tier)	**$0.00** (Kaggle Free Tier)

4. Google Drive Submission Folder Structure

Per Techfest Page 5 guidelines, all submission materials are organized inside the `submission/` directory:

1. `MASTER_SUBMISSION_DOCUMENT.pdf`: This master document compiling all supplementary links and materials.
2. `Technical_Abstract_Topoloka.pdf`: 4-page formal technical abstract.
3. `topoloka_demo.mp4`: Merged 1080p demonstration video (Metro walkthrough, Point cloud inspection, Studio Room HUD controls).
4. `Proof_of_Concept_Web_Viewer_Link.txt`: Direct URL to the live hosted WebGL twin.
5. `Demonstration_Video_Link.txt`: Direct URL to the video stream and download.
6. `Code_Repository_Link.txt`: Direct URL to the GitHub repository.